

Kinematics — Motion, Vectors and Limits

Lecture Notes

2026

Introduction

Kinematics is the geometry of motion — it describes *how* objects move without asking *why*. The language is calculus: velocity is the derivative of position, acceleration is the derivative of velocity. Algebra gives you the results; calculus tells you why they are true.

Reference: Halliday, Resnick & Krane, *Physics* Vol. 1, Chapters 2–4. All formulas here follow from two definitions: $\vec{v} = d\vec{r}/dt$ and $\vec{a} = d\vec{v}/dt$.

1. Frame of Reference

All motion is measured relative to a chosen **inertial frame** — a coordinate system at rest or moving at constant velocity. The same event has different descriptions in different frames; the laws of motion hold in all inertial frames.

There is no absolute rest. When you say a ball moves at 10 m/s, you mean relative to Earth. A passenger on a train sees a different velocity. Fix your frame before writing any equation.

2. Scalars and Vectors

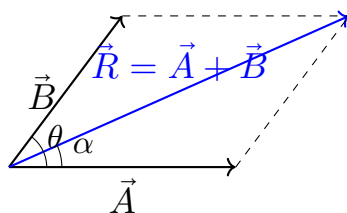
A **scalar** has magnitude only: speed v , distance d , time t .

A **vector** has magnitude and direction: position $\vec{r}$, velocity $\vec{v}$, acceleration $\vec{a}$.

Vector Addition — Parallelogram Law

For $\vec{A}$ and $\vec{B}$ at angle θ :

$$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB \cos \theta} \quad \tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$



Special cases: $\theta = 0^\circ \Rightarrow R = A + B$; $\theta = 180^\circ \Rightarrow R = |A - B|$; $\theta = 90^\circ \Rightarrow R = \sqrt{A^2 + B^2}$.

3. Equations of Motion (Constant Acceleration)

Starting from $\vec{a} = d\vec{v}/dt = \text{const}$, integrating once gives $\vec{v}$, integrating again gives $\vec{r}$:

$$\frac{dv}{dt} = a \xrightarrow{\int_0^t dt} v = u + at$$

Velocity changes linearly with time; the slope is acceleration.

$$v = \frac{ds}{dt} = u + at \xrightarrow{\int_0^t dt} s = ut + \frac{1}{2}at^2$$

Displacement is the area under the v - t graph — a trapezoid, hence the $\frac{1}{2}$ factor.

$$v^2 = u^2 + 2as$$

Eliminate t : energy-like relation between velocity and displacement.

$$s_n = u + \frac{a}{2}(2n - 1)$$

Distance in the n th second: the difference $s(n) - s(n - 1)$, which is an arithmetic progression with common difference a .

4. Motion Under Gravity

Free fall is constant acceleration with $\vec{a} = -g\hat{j}$, $g \approx 9.8 \text{ m/s}^2$:

$$\frac{d^2y}{dt^2} = -g \xRightarrow{\text{integrate twice}} y = ut - \frac{1}{2}gt^2$$

The minus sign is the frame choice: upward positive.

5. Projectile Motion

Decompose into independent axes. Horizontal: $a_x = 0$. Vertical: $a_y = -g$.

$$u_x = u \cos \theta, \quad u_y = u \sin \theta$$

Quantity	Formula	Calculus origin
Time of flight	$T = 2u \sin \theta / g$	$y(T) = 0$, solve quadratic in t
Max height	$H = u^2 \sin^2 \theta / (2g)$	$dy/dt = 0$ at apex
Range	$R = u^2 \sin 2\theta / g$	$x(T)$, use double-angle identity

Trajectory (eliminate t from $x = u_x t$, $y = u_y t - \frac{1}{2}gt^2$):

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

This is a downward parabola: $y = Ax + Bx^2$ with $B < 0$.

R is maximum at $\theta = 45^\circ$ since $\sin 2\theta$ peaks there. Complementary angles θ and $90^\circ - \theta$ give equal range.

6. Solved Problem — Angular Momentum of a Projectile

Problem. A ball is launched from origin P with speed u at angle α . At time t , the position vector $\vec{r}$ from P makes angle θ with the horizontal. Find the angular momentum $\vec{L}$ about P .

Solution.

Position and velocity at time t :

$$\vec{r} = (u \cos \alpha) t \hat{i} + \left[(u \sin \alpha) t - \frac{1}{2} g t^2 \right] \hat{j}$$

$$\vec{v} = (u \cos \alpha) \hat{i} + (u \sin \alpha - g t) \hat{j}$$

Angular momentum $\vec{L} = m(\vec{r} \times \vec{v})$. Only the $\hat{k}$ component survives in 2D:

$$\vec{r} \times \vec{v} = (x v_y - y v_x) \hat{k}$$

Expanding:

$$\begin{aligned} x v_y - y v_x &= (u \cos \alpha) t (u \sin \alpha - g t) - \left[(u \sin \alpha) t - \frac{1}{2} g t^2 \right] (u \cos \alpha) \\ &= (u \cos \alpha) \left[t(u \sin \alpha - g t) - t u \sin \alpha + \frac{1}{2} g t^2 \right] \\ &= (u \cos \alpha) \left[-g t^2 + \frac{1}{2} g t^2 \right] = -\frac{1}{2} g t^2 u \cos \alpha \end{aligned}$$

Since $\vec{r}$ makes angle θ with horizontal, $x = |\vec{r}| \cos \theta$ and $u \cos \alpha = x/t = |\vec{r}| \cos \theta/t$:

$$\boxed{|\vec{L}| = \frac{1}{2} m g t |\vec{r}| \cos \theta}$$

directed into the page. The angular momentum grows linearly with t because gravity exerts a constant torque $\tau = m g x = m g |\vec{r}| \cos \theta$ about P , and $d\vec{L}/dt = \vec{\tau}$.

7. The Fly and the Train — Zeno Resolved by Calculus

Setup. Initial gap D . Fly moves at $v = V/2$ toward train; train moves at V toward fly. Closing speed = $v + V = 3V/2$.

Zeno's series approach (dichotomy: halve the gap at each step):

$$t_{total} = \sum_{n=1}^{\infty} \frac{D/2^n}{v + V} = \frac{D}{v + V} \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{D}{v + V} \cdot 1 = \frac{D}{3V/2} = \frac{2D}{3V}$$

The sum $\sum 1/2^n = 1$ (geometric series, ratio $r = 1/2 < 1$). The series converges to a **finite** time. The fly hits the train at $t^* = 2D/(3V)$.

Calculus approach (the definitive one):

Let $x(t)$ be the gap. It closes at constant rate:

$$\frac{dx}{dt} = -(v + V) = -\frac{3V}{2}, \quad x(0) = D$$

$$x(t) = D - \frac{3V}{2}t$$

Set $x(t^*) = 0$:

$$t^* = \frac{2D}{3V}$$

Zeno's infinite subdivision describes the same process in infinitely many steps. The geometric series sums to a finite limit because the steps shrink faster than they accumulate. Calculus bypasses the infinite steps entirely by treating the rate of change as a continuous quantity — dx/dt is defined precisely by the limit $\lim_{\Delta t \rightarrow 0} \Delta x / \Delta t$. Both approaches agree: the fly hits the train. The event is absolute; only the method of counting steps is relative.

Key Formulae

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

$$R = \frac{u^2 \sin 2\theta}{g} \quad H = \frac{u^2 \sin^2 \theta}{2g} \quad T = \frac{2u \sin \theta}{g}$$

$$|\vec{L}| = \frac{1}{2}mgt|\vec{r}| \cos \theta \quad \frac{d\vec{L}}{dt} = \vec{\tau} \quad a_c = \frac{v^2}{r} = r\omega^2$$